sas innovate 2026

Better (Prompt) Engineering: Leveraging ChatGPT to Get Started with SAS Coding

Lincoln H. Groves, PhD | SAS Academic Programs
www.linkedin.com/in/lincoln-h-groves/

Copyright © SAS Institute Inc. All rights reserved.

Pulse Check

Thinking about ChatGPT + other LLMs…

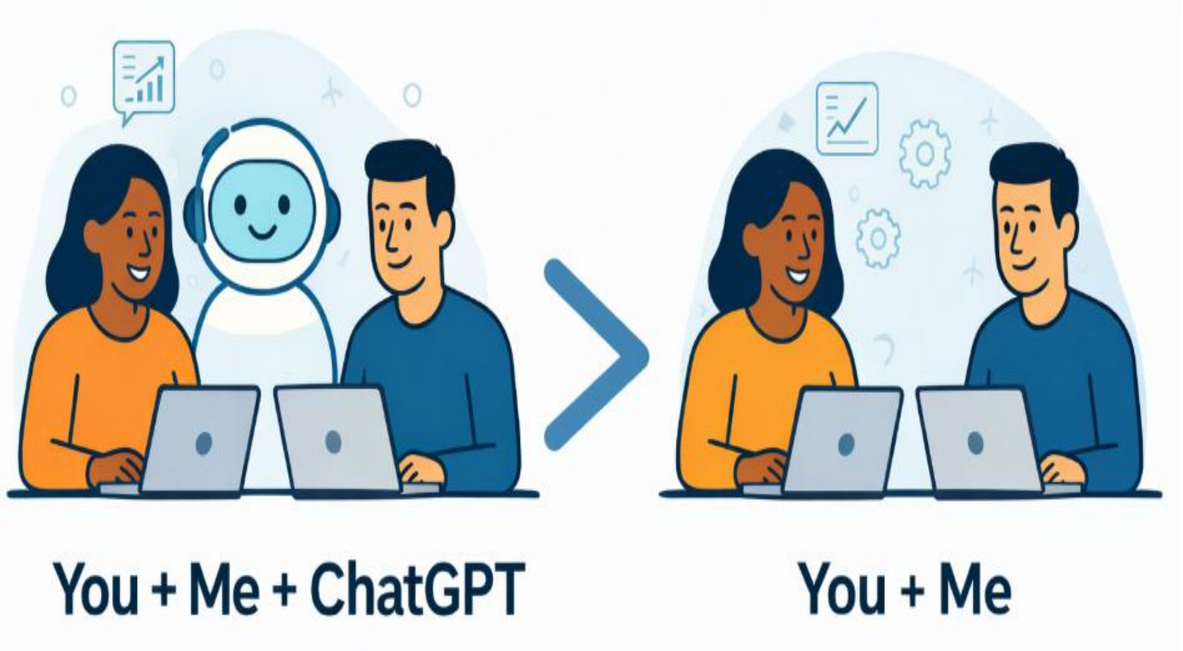
- Who is using LLMs weekly?
  - Daily?
- Be honest: who thinks AI is still a bit of a black box?

Copyright © SAS Institute Inc. All rights reserved.

Why This Matters Now

LLMs are already part of the workflow

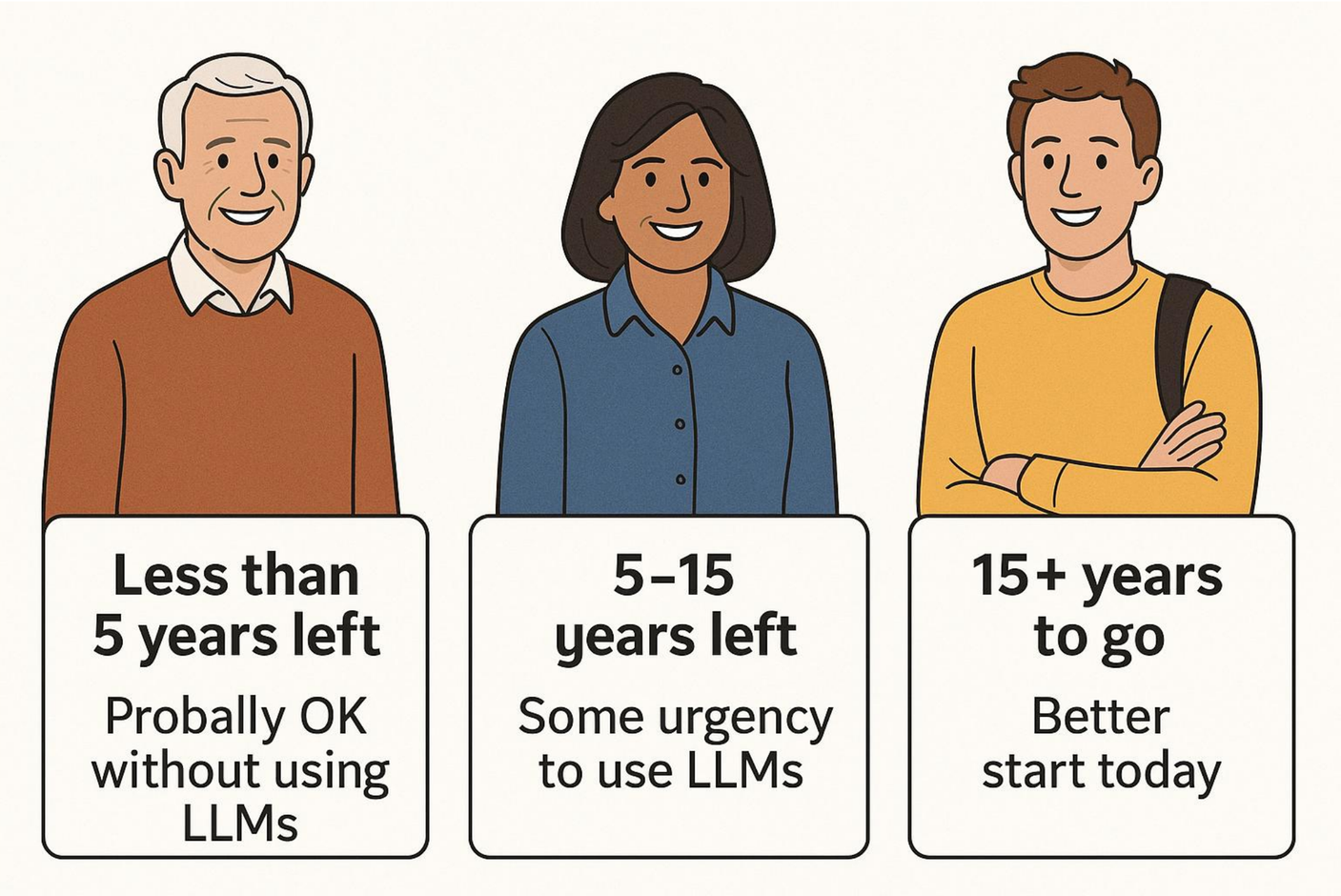
- AI is changing entry-level work
- Expectations are rising for others
- The advantage is learning to work with these tools



Copyright © SAS Institute Inc. All rights reserved.

My (not so subtle) advice

Make friends with the LLMs!



Copyright © SAS Institute Inc. All rights reserved.

What ChatGPT Actually Is

It doesn’t know — it predicts.

- Large Language Model
- Trained on text + code
- Predicts patterns
- Not a database
- Not an expert
- Not a person

ChatGPT

Copyright © SAS Institute Inc. All rights reserved.

The Real Role of ChatGPT

ChatGPT is not a coder. It's a collaborator.

- Start faster
- Understand more
- Work better

You still provide judgment!

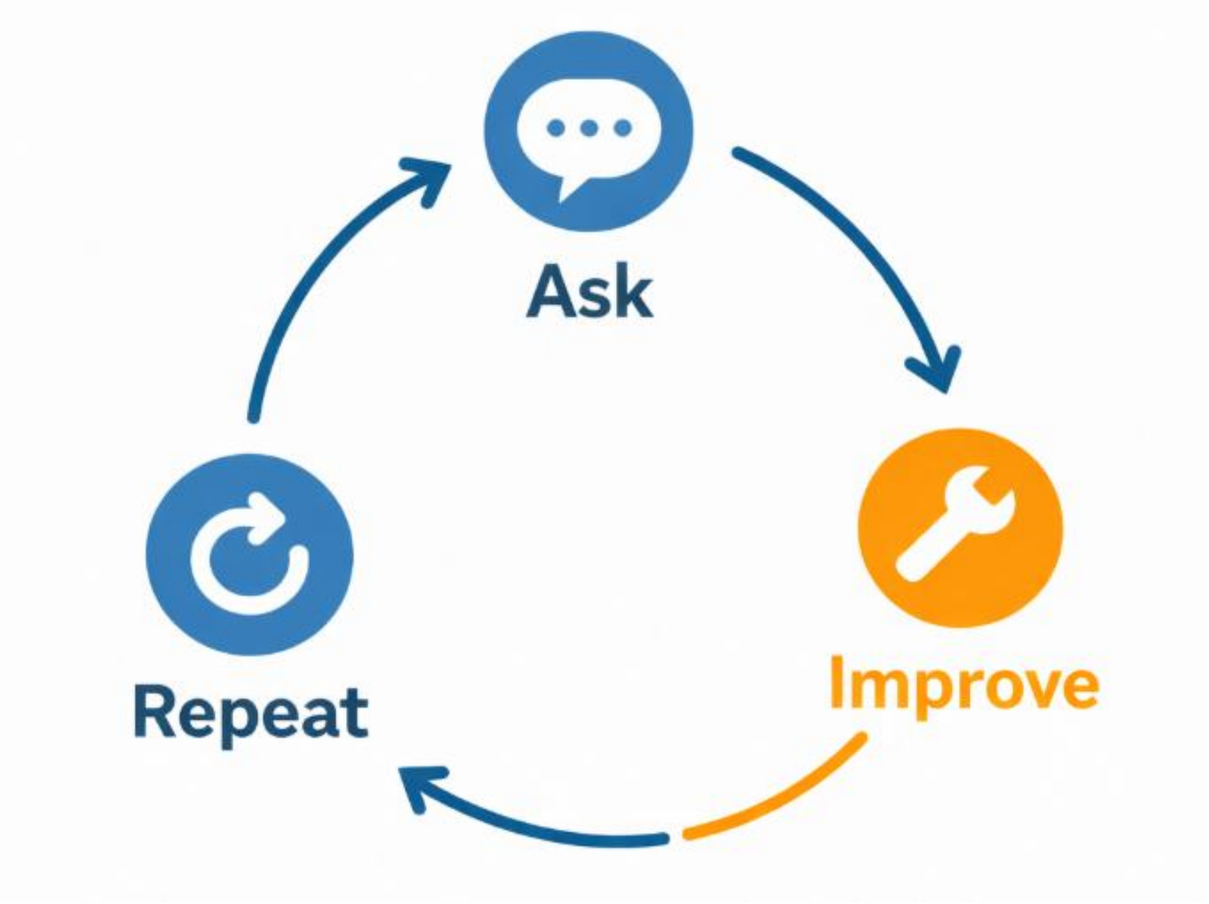
Copyright © SAS Institute Inc. All rights reserved.

A Simple Prompt Framework

Iteration is expected — not failure

Better prompts usually include:

- Context — what are you doing?
- Task — what do you want?
- Output — what should the answer look like?



Copyright © SAS Institute Inc. All rights reserved.

What Lies Ahead in this Workshop

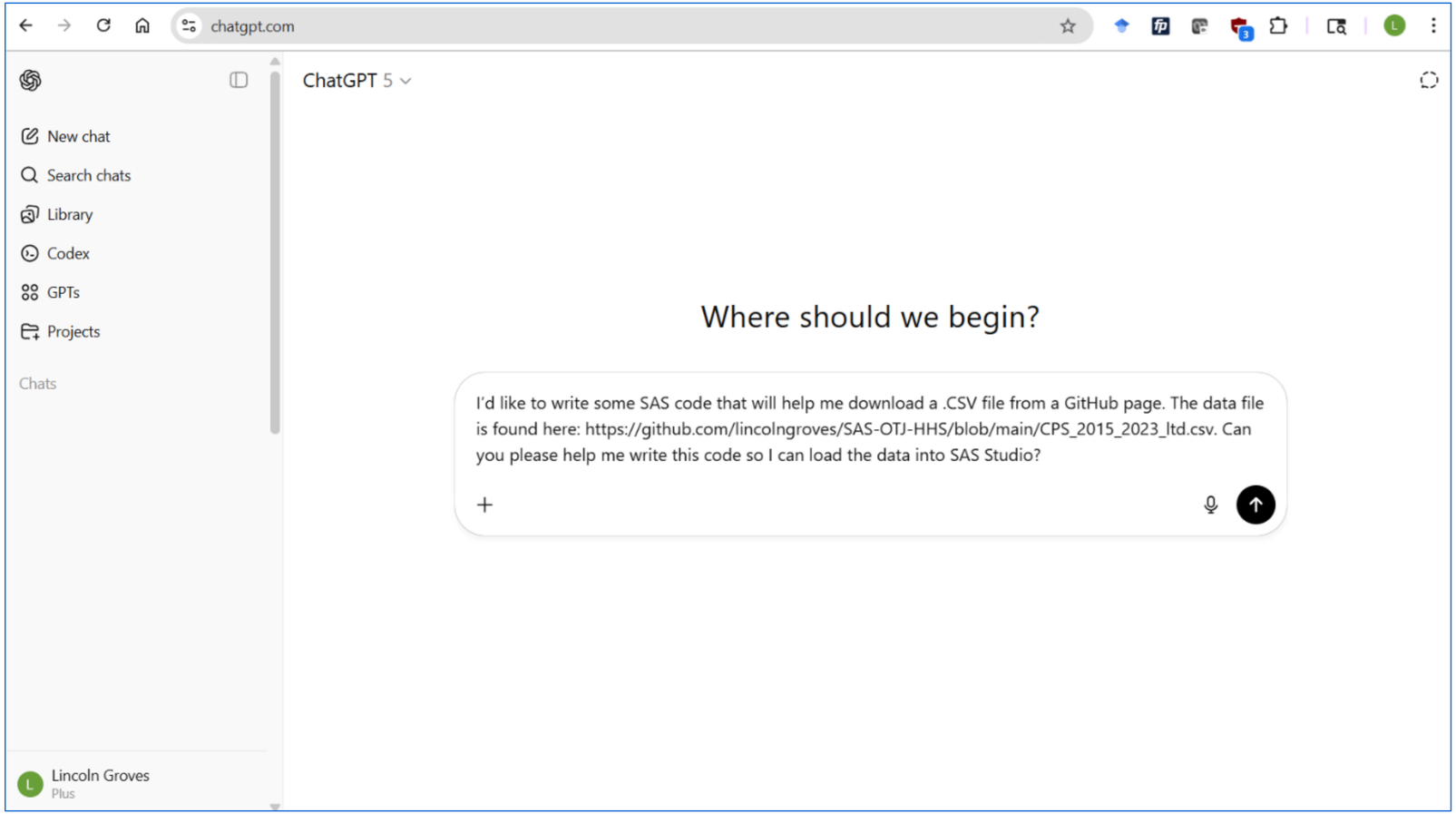
Ways ChatGPT can improve your workflow

- Start faster
- Understand more
- Work better

Copyright © SAS Institute Inc. All rights reserved.

Act 1 | Start Faster

Generating a first draft of code



Go from not knowing where to start… to starting… in seconds

Copyright © SAS Institute Inc. All rights reserved.

Act 2 | Understand More

From generating code to interpreting code

Demystify the mystified.

Or just figure out what you did months ago.

I'm struggling to understand some complex proc SQL code in SAS. Allegedly, this code takes a very long data set and collapses the data to a much small number of observations, while creating a bunch of new variables. Can you help explain it to me like I'm a 10 year old? Here is the code:

```
************************************************************ By State ;
proc sql;
   create  table hhs.covid_labor_supply as
   select   distinct state_fip, state_name,
         year(yearquarter) as Year format 9.,

/*****************************************************************  Labor Force Status | All  */
         sum( ( unemp=1 ) * WTFINL )                                  / sum( ( in_LF=1 ) *     WTFINL )
as UE_Women          label="Unemployment Rate"  format percent9.1       ,
```

ChatGPT can make mistakes. Check important info.

Copyright © SAS Institute Inc. All rights reserved.

Act 3 | Work Better

Move from getting code to being a better analyst/teammate.

- Refine outputs
- Explore results
- Document your work
- Organize like a professional coder

I'm proud of this code that I just wrote. But I'm tired and don't feel like documenting what I did properly. Can you help insert some comments into this code so that future me can understand what I did? Here is the code:

```
*****************************************************  By State ;
proc sql;
   create  table hhs.covid_labor_supply as
   select  distinct state_fip, state_name,
         year(yearquarter) as Year format 9.,

/*****************************************************************  Labor Force Status | All  */
         sum( ( unemp=1 ) * WTFINL )                                  / sum( ( in_LF=1 ) *     WTFINL )
as UE_Women          label="Unemployment Rate"  format percent9.1      ,
```

+

ChatGPT can make mistakes. Check important info.

Copyright © SAS Institute Inc. All rights reserved.

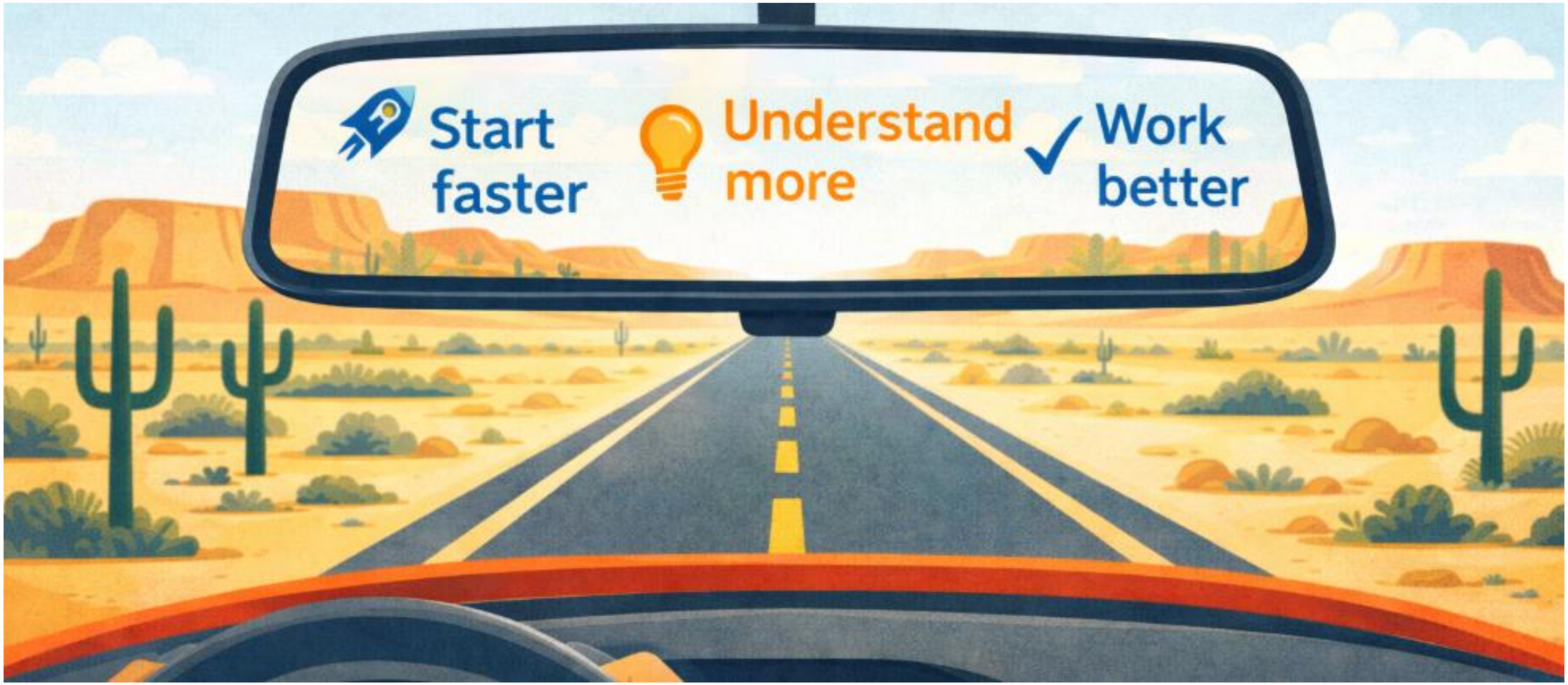
sas innovate 2026

Demo Time!

Copyright © SAS Institute Inc. All rights reserved.

What Just Happened (and Why It Matters)

Returning to the 3-part framework



Copyright © SAS Institute Inc. All rights reserved.

Lessons Learned

And how to apply them tomorrow....

Co-create, don't copy

Better prompts → better output

Move faster, don't think less

ChatGPT complements SAS expertise

Copyright © SAS Institute Inc. All rights reserved.

“

The winners won’t just be strong coders.

They’ll be strong collaborators.

With humans… and with machines.

Copyright © SAS Institute Inc. All rights reserved.

Resources

Learn More + Keep Exploring

Copyright © SAS Institute Inc. All rights reserved.

GitHub Links

Repo for this workshop

- https://github.com/lincolngroves/BetterPromptEngineering/
- All the steps, in one PDF

THE POWER TO KNOW.

Better (Prompt) Engineering:

Leveraging ChatGPT to get started with SAS Coding

SAS Guided Demo

Purpose

This SAS Guided Demo introduces programmers to the practical use of large language models (LLMs), such as ChatGPT, as a partner in code development. We walk through three main cases where LLMs can be particularly useful for those learning new code: (1) in generating a first draft of code, which gets you 80-90% of the way to a working solution, (2) understanding complex code written by someone else, and (3) supporting code documentation, to make your work clearer and easier to maintain. Moreover, we start with best practices on how to interact with LLMs to optimize the output generated.

Software

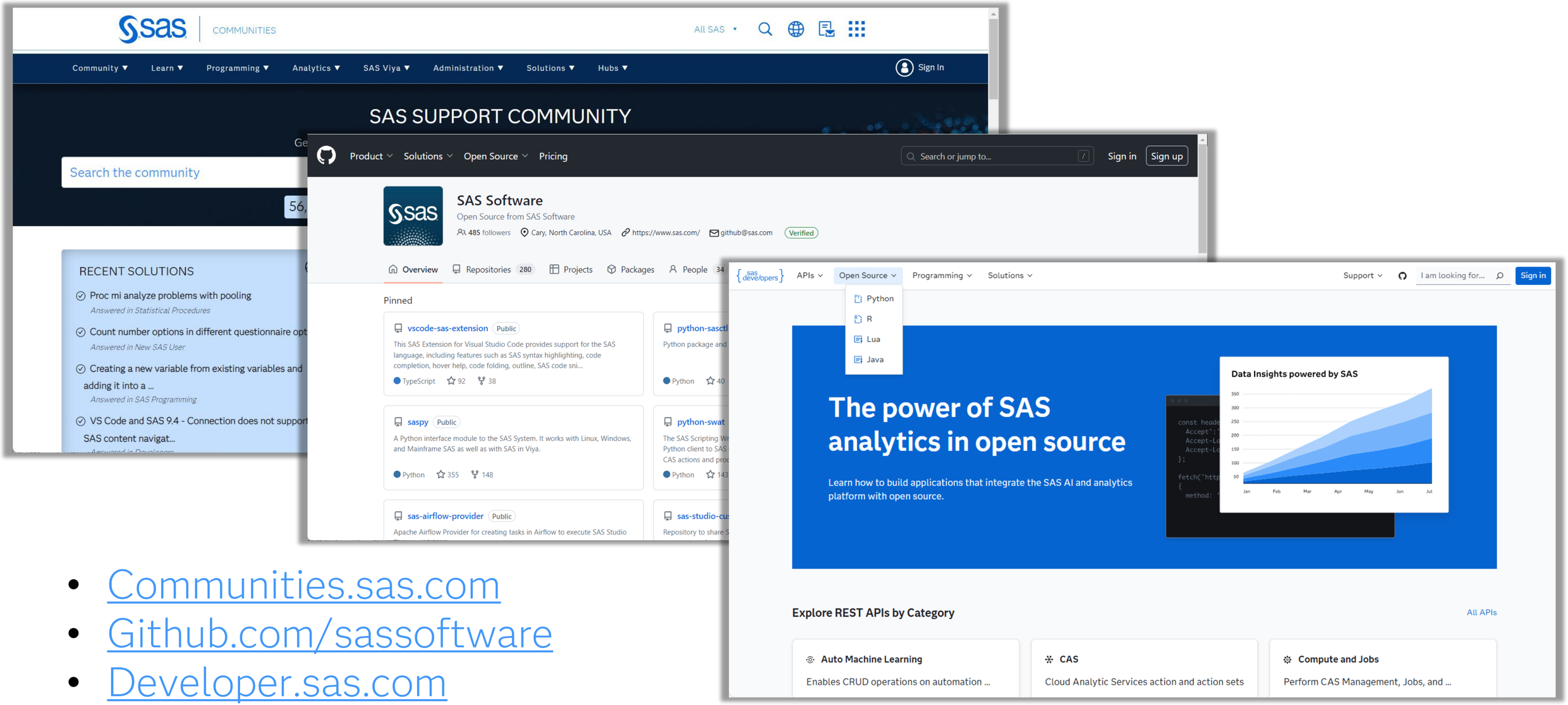
This SAS Guided Demo uses SAS Viya for Learners, version 2025.03LTS, and the free version of ChatGPT.

Prerequisites

While not mandatory, it is useful if participants have some exposure to SAS coding in SAS Model Studio and in working with ChatGPT.

Copyright © SAS Institute Inc. All rights reserved.

Other great resources…



- Communities.sas.com
- Github.com/sassoftware
- Developer.sas.com

Copyright © SAS Institute Inc. All rights reserved.

Fall 2025 SAS Communities Series:

👉 You + Me + ChatGPT Make Three: Using LLMs to get started with SAS Programming | https://communities.sas.com/t5/SAS-Communities-Library/You-Me-ChatGPT-Make-Three-Using-LLMs-to-get-started-with-SAS/ta-p/977749

💭 Use ChatGPT to help with the first iteration of code https://communities.sas.com/t5/SAS-Communities-Library/Use-ChatGPT-to-help-with-the-first-iteration-of-code/ta-p/977785

🧠 Use ChatGPT to help understand complex code (yours or someone else's) | https://communities.sas.com/t5/SAS-Communities-Library/Use-ChatGPT-to-help-understand-complex-code/ta-p/977786

📝 Use ChatGPT to document your code today — so future-you says thank you | https://communities.sas.com/t5/SAS-Communities-Library/Use-ChatGPT-to-document-your-code-today/ta-p/977787

Copyright © SAS Institute Inc. All rights reserved.

sas innovate 2026

Thank you!

Let’s keep the conversation going — connect with me on LinkedIn!
https://www.linkedin.com/in/lincoln-h-groves/

Copyright © SAS Institute Inc. All rights reserved.